PRACTICAL 5

Links, Archives and Compression

Aim

To create hard/soft links and manage compressed and archived files.

A. Hard Link

Create a file:

echo "TYIT Linux" > original.txt

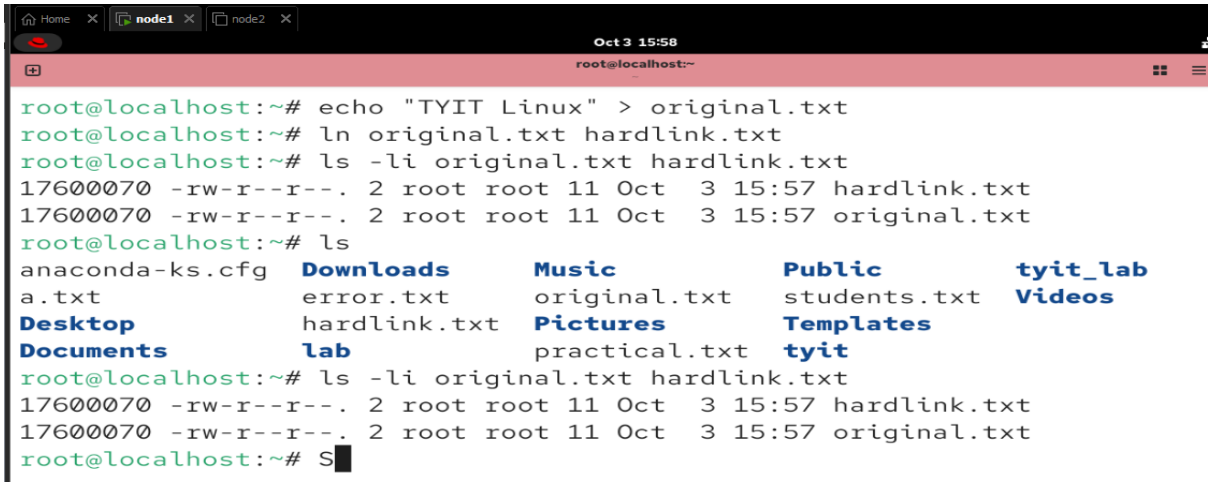
Create hard link:

ln original.txt hardlink.txt

Check inode:

ls -li original.txt hardlink.txt

Both files should have the same inode number.



B. Soft Link

ln -s original.txt softlink.txt

Check:

ls -l softlink.txt

Expected:

softlink.txt -> original.txt

```
root@localhost:~# ln -s original.txt softlink.txt
root@localhost:~# ls -l softlink.txt
lrwxrwxrwx. 1 root root 12 Oct  3 15:59 softlink.txt -> original.txt
root@localhost:~#
```

C. tar

Create archive:

tar -cvf tyit.tar original.txt hardlink.txt

List archive:

tar -tvf tyit.tar

Extract:

mkdir extracted

tar -xvf tyit.tar -C extracted

D. gzip

gzip original.txt

Check:

ls

Uncompress:

gunzip original.txt.gz

E. bzip2

bzip2 original.txt

Uncompress:

bunzip2 original.txt.bz2

F. star

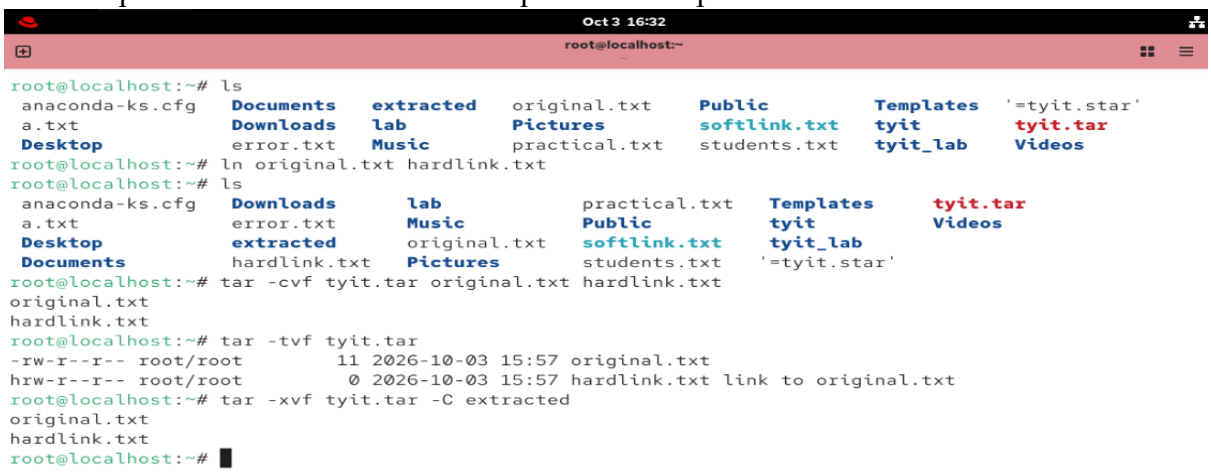
If star is installed:

star -c -f=tyit.star original.txt

List:

star -t -f=tyit.star

If star is not installed in your RHEL installation, install the package through the configured RHEL repositories or demonstrate the equivalent tar operation.



```
root@localhost:~# tar -c -f=tyit.star original.txt
root@localhost:~# star -t -f=tyit.star
bash: star: command not found...
Similar command is: 'tar'
root@localhost:~# tar -t -f=tyit.star
original.txt
root@localhost:~#
```

```
root@localhost:~# gzip original.txt
root@localhost:~# ls
anaconda-ks.cfg  Downloads  Music             Public        tyit
a.txt            error.txt  original.txt.gz   softlink.txt  tyit_lab
Desktop          extracted  Pictures          students.txt  tyit.tar
Documents        lab        practical.txt     Templates     Videos
root@localhost:~# gunzip original.txt.gz
root@localhost:~# ls
anaconda-ks.cfg  Downloads  Music             Public        tyit
a.txt            error.txt  original.txt      softlink.txt  tyit_lab
Desktop          extracted  Pictures          students.txt  tyit.tar
Documents        lab        practical.txt     Templates     Videos
root@localhost:~#
```

PRACTICAL 6
File Ownership, Permissions and ACL

Aim
To manage Linux ownership, standard permissions, special permissions and ACLs.

A. View permissions
ls -l
Example:
-rw-r--r-- 1 student student 15 Sep 20 10:20 file.txt
Permission format:

```
- rw- r-- r--
  |  |  |
  |  |  +--- Others
  |  +------- Group
  +---------- Owner
```

B. chmod
Create file:
touch permission.txt
Set permissions:
chmod 755 permission.txt
Check:
ls -l permission.txt

Numeric Permissions

Number	Permission
0	---
1	--x
2	-w-
3	-wx
4	r--
5	r-x
6	rw-
7	rwx

Example:
chmod 644 file.txt

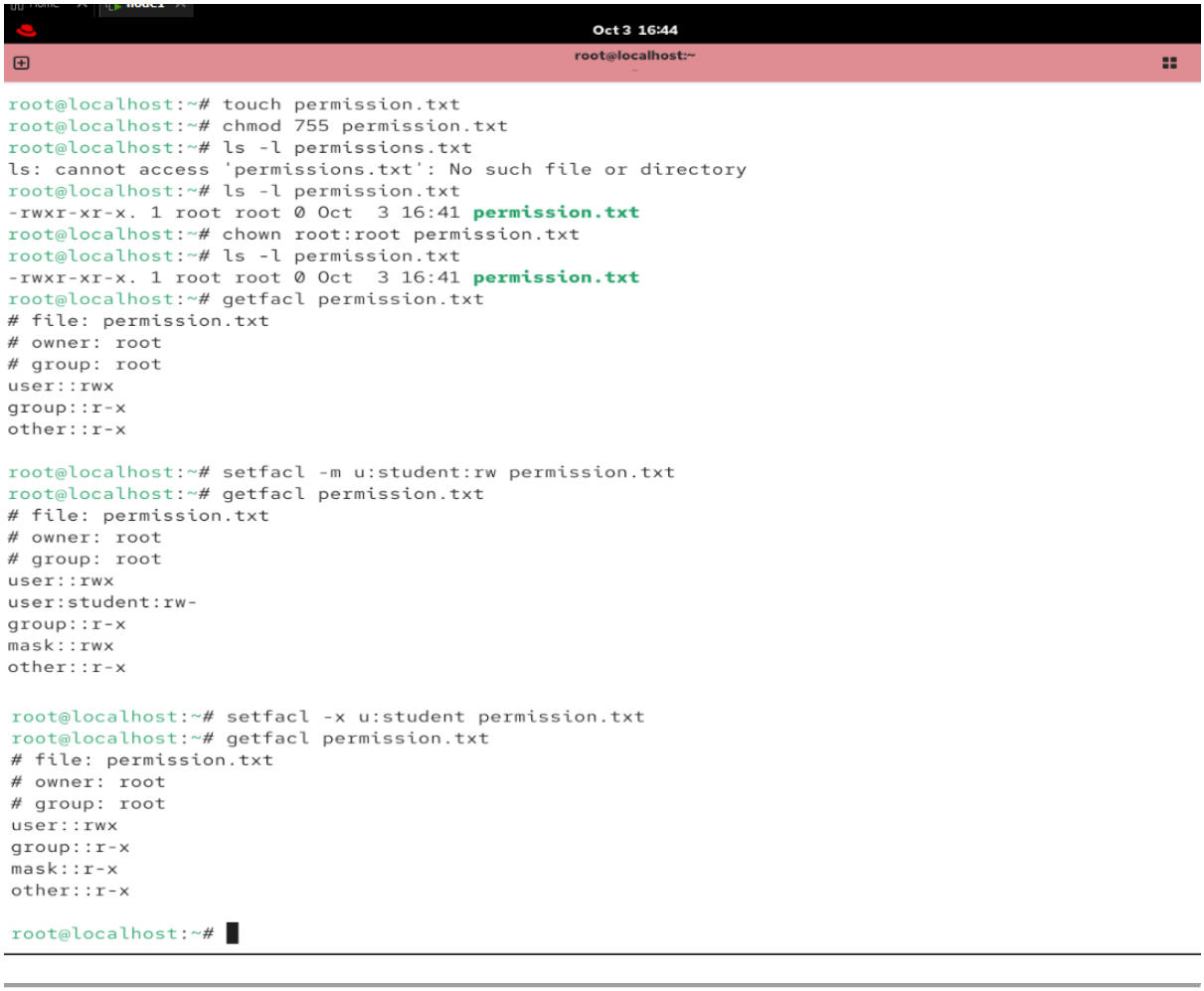
C. chown
sudo chown root:root permission.txt
Check:
ls -l permission.txt

D. umask
umask
Temporarily set:

umask 027
Create:
touch umaskfile
Check:
ls -l umaskfile

E. set-GID Directory
Create group:
sudo groupadd project
Create directory:
sudo mkdir /project
Change group:
sudo chgrp project /project
Set set-GID:
sudo chmod 2775 /project
Verify:
ls -ld /project
The group directory should show s in the group permission position.

F. ACL
Check ACL:
getfacl permission.txt
Set ACL:
sudo setfacl -m u:student:rw permission.txt
Check:
getfacl permission.txt
Remove ACL:
sudo setfacl -x u:student permission.txt


Oct 3 16:44
root@localhost:~
root@localhost:~# touch permission.txt
root@localhost:~# chmod 755 permission.txt
root@localhost:~# ls -l permissions.txt
ls: cannot access 'permissions.txt': No such file or directory
root@localhost:~# ls -l permission.txt
-rwxr-xr-x. 1 root root 0 Oct 3 16:41 permission.txt
root@localhost:~# chown root:root permission.txt
root@localhost:~# ls -l permission.txt
-rwxr-xr-x. 1 root root 0 Oct 3 16:41 permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
group::r-x
other::r-x
root@localhost:~# setfacl -m u:student:rw permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
user:student:rw-
group::r-x
mask::rwx
other::r-x
root@localhost:~# setfacl -x u:student permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
group::r-x
mask::r-x
other::r-x
root@localhost:~#